Tips til rapportskriving for prosjekt- og masteroppgave

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Oversikt over forelesningen

- Motivasjon og hensikt

- Rapportskriving:
  - Rapportstruktur
  - Råd til ulike kapitler
  - Generelle råd
- Selvplagiering
- Bruk av KI-verktøy

Motivasjon og hensikt

- Komme raskt i gang med skriveprosessen
- Lette arbeidet med å skrive rapport
- Kvalitet på rapporten endrer lett karakteren – sensor ser bare denne

- Istedenfor å gi råd til hver gruppe, gi et felles basistilbud
- Råd gjelder for både prosjekt- og masteroppgaven

- Lærer allikevel best ved å se på eksisterende rapporter til «liknende» prosjekt
- …samt å høre på veileder!

Prosjekt- og masteroppgave: Hva er det?

- Selvstendig forskningsprosjekt gjennomført av studenten under veiledning av faglig ansatte
- Denne høsten: Prosjektoppgave, neste semester: Masteroppgave
  - Prosjektoppgaven og masteroppgaven er to separate vitenskapelige arbeider, derfor settes det to ulike karakterer.
  - Vanligvis velger studentene å fortsette med samme tema i masteroppgaven, for eksempel ved å snevre inn eller utvide problemstillingen.
  - Likevel må oppgavene være tilstrekkelig forskjellige!
- Forskjell mellom masteroppgave og prosjektoppgave:

Prosjektoppgave:

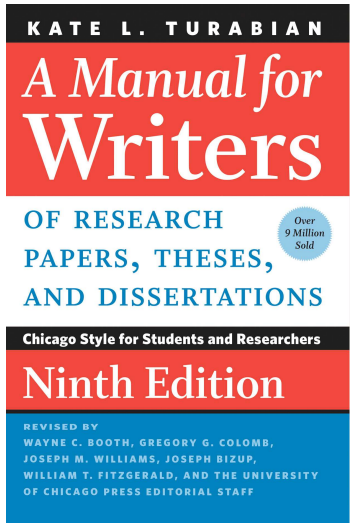
- Selvstendig forskningsprosjekt
- Trenger ikke å være banebrytende forskning
- Studentene må demonstrere at de er:
  - kjent med litteraturen
  - kjent med bransjens særtrekk
  - i stand til å anvende metoder studert i undervisningen på et reelt problem
- Et første forsøk på modellering
  - Noe nytenkning
- Ikke offentlig publisert arbeid

Masteroppgave:

- Et mer dyptgående og omfattende forskningsprosjekt
- Nyskapende og originale bidrag til fagfeltet
- Studentene må demonstrere at de er i stand til å løse problemer utover det som er studert i undervisningen
- Mye mer fokusert på et spesifikt forskningsspørsmål,
  - F.eks. mindre plass til generell bransjebakgrunn eller litteratur som ikke er direkte nødvendig for å forstå problemet
  - Mer konsist
- Offentlig publisert (konfidensialitet?)

Akademisk skriving

- Den beste måten å lære å skrive akademisk på en god måte er å lese (mange) veldig gode artikler …
- Her får du likevel noen retningslinjer for akademisk skriving:
  - https://innsida.ntnu.no/oppgaveskriving
- Og også litt mer informasjon om masteroppgaven:
  - https://i.ntnu.no/masteroppgave
- Noen tips fra personlig erfaring på de neste lysbildene



Generelt

- Dere har et *budskap* som skal formidles
- Dere kan ikke se ansiktene til publikum (dvs. leseren) og de kan ikke stille dere oppklaringsspørsmål
- Dere har kun ett forsøk – det bør derfor være bra!
- Alt innhold må være relevant for budskapet
- Strukturen på teksten skal gjøre budskapet lettere å forstå for leseren

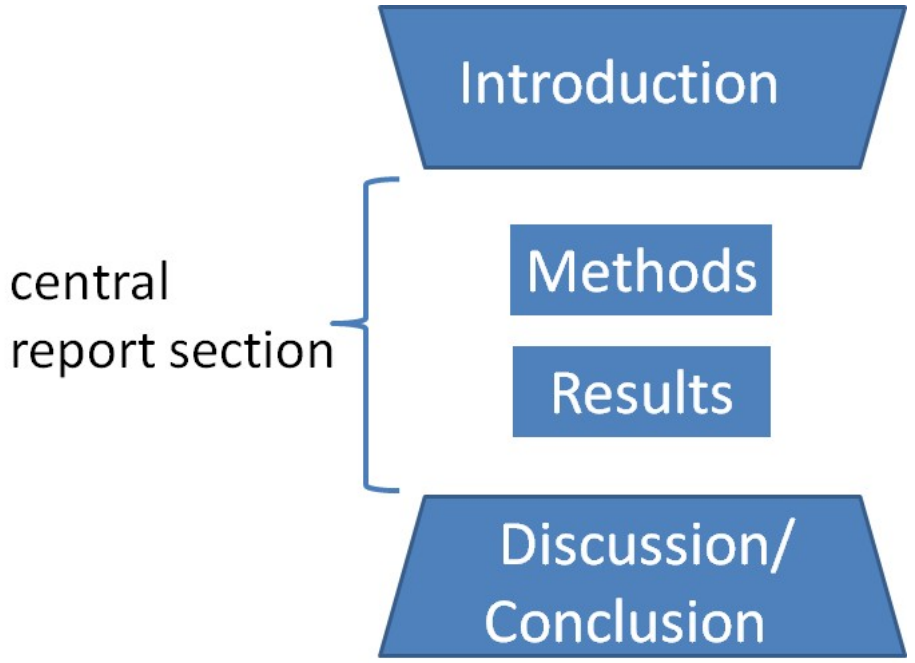
- Forsøk å se over egen tekst med leserens øyne («glem» alt dere har lært underveis i arbeidet og se om dere fortsatt forstår budskapet)

Hvem skriver dere til?

- Publikum:
  - Skriv til en medstudent med AØO fordypning. Beskriv teori som læres våren i 4. klasse og høsten i 5. klasse. Tidligere teori antas kjent
  - Dette betyr f.eks. at dere ikke behøver å forklare Simplex-algoritmen eller branch-and-bound
  - Dere kan derimot ikke forvente at leseren kan veldig mye om problemet/sektoren som dere jobber med – dette bør derfor oftest forklares godt

Rapportstruktur – overordnet

IMRaD



https://en.wikipedia.org/wiki/IMRAD

Rapportstruktur – detaljer

- Flere mulige varianter – Ingen fasit
- Avhenger mye av problemstilling og type prosjekt (praktisk/teoretisk – optimering/økonomi)
- Fornuftig å tenke på tidlig, men vær forberedt på å endre når dere har skrevet større deler av rapporten
- Tipsene i denne presentasjonen er nok aller mest tilpasset typiske optimeringsoppgaver (uansett, hør først og fremst på veileder)

To mulige rapportstrukturer

Alternative 1:

Preface
Summary
Contents/List of figures/List of tables/
(Terminologi list/Abbreviations)
1. Introduction
2. Background
3. Related literature
4. Problem description
5. Mathematical model
6. Solution method
7. Implementation
8. Computational results
9. Concluding remarks
10. Future research
Appendices
Reference list

Alternative 2:

Preface
Summary
Contents/List of figures/List of tables/
(Terminologi list/Abbreviations)
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2. Problem description
3. Related literature
4. Mathematical model
5. Solution method
6. Computational study
7. Concluding remarks and future research
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…. eller en kombinasjon av disse
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NTNU

Struktur: hva hører hjemme hvor?

- Strukturer innholdet slik at leseren så enkelt som mulig kan følge og forstå «historiefortellingen» og budskapet

- Vær nøye mtp. hvilke informasjon som skal gå hvor, f.eks.
  - Ikke diskuter resultater i metodekapitlet
  - Ikke beskriv data (tall) i modell- eller metodekapittel

Råd til ulike kapitler – Tittel først

- Ikke veldig viktig for prosjektoppgave
- Tittel på masteroppgave står på vitnemål
  - Tenk over hvordan dere vil markedsføre dere
  - Gjem «optimal» tittel til masterrapporten
  - Kombiner fagområde og anvendelsesområde
- Unngå repetisjon: "The stochastic vehicle routing problem with stochastic demands"
- Unngå forkortelser: "A new EOQ model"
- Kan også bruke undertitler (f.eks. til å poengtere løsningsmetode)

- Eksempler:
  - A new optimization-based algorithm for a maritime inventory routing problem
  - Supply chain optimization in the Liquefied Natural Gas business
  - The static ridesharing routing problem with flexible candidate locations: an adaptive large neighborhood search heuristic

Lengde på rapport

- Ingen absolutte krav
- Anbefaler følgende:
  - Maks 80 sider på prosjekt og 100 på master (ekskl. appendiks)

- Uansett, vær konsis - ingen ønsker å lese mer enn nødvendig
- Det bør være en sammenheng mellom hvor viktig noe er, og hvor mye plass det tar i rapporten
- Skriv om det som fungerte, ikke alt dere har prøvd ut på veien. Trenger **ikke** hele historien
  - Først prøvde vi A, men det fungerte ikke så vi gjorde B, men da ble det enda dårligere. Til slutt fant vi ut at C var den beste måten å gjøre det på ← **Ikke slik**

Preface

- Kan være kort
- Oppgi navn og kode for fordypningsprosjektet/master fordypning
- For master: Referer til prosjekt og si at dette er en videreføring
  - Kan også bruke dette til å deklarere selvplagiering (mer senere)
- Takk til industripartner
- Takk til veileder(e)

Abstract/Summary (1:2)

- Sjefsvarianten
- Max 1 side!
- Skal være et sammendrag av hele rapporten
- I master skal en ha både engelsk og norsk variant

Abstract/Summary (2:2)
Timeglasstruktur (IMRaD)

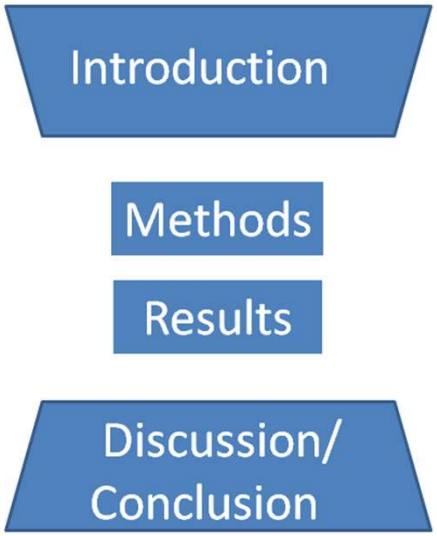
Presenter bakgrunnsinformasjon/motivasjon for studien

Identifiser kunnskapshull i forskningslitteraturen

Presenter mål og metode for studien

Gi en oppsummering av viktigste resultater/bidrag

Beskriv de bredere implikasjonene av disse resultatene



Lister

- Inkluder Contents, List of Figures, List of Tables
- Anbefaler automatisk generering
- Sørg for at det stemmer, hvis ikke automatisk generert
- For noen passer det også med terminologiliste. Det kan være forkortelser (MIP, RMP, VRPTWPC, …) eller forklaring av fagterminologi fra anvendelsesområde (nedetid, ønsketurnus,….)

Eksempel terminologiliste

Dictionary

Abbreviations

LS	–	Left side
NA	–	Not applicable
RS	–	Right side

Industry Terminology

Calcine	–	To convert to calx by heating or burning
Calx	–	The oxide substance that remains after metals have been thoroughly roasted
HC	–	High Carbon
LC	–	Low Carbon
MC	–	Medium Carbon
MOR	–	Manganese Oxygen Refining
Reactant	–	Any substance that undergoes a chemical change in a given reaction
Redox	–	Reduction-oxidisation
Resultant	–	The result from the combination of two or more reactants
Tonne	–	Metric tonne, 1000kg

Optimisation Terminology

BLP	–	Bilinear Program
GOP	–	Global Optimisation Algorithm
LP	–	Linear Program
MAMP	–	Manganese Alloy Multi-plant Production problem
MDT	–	Multiparametric Disaggregation Technique
MILP	–	Mixed Integer Linear Program
MINLP	–	Mixed Integer Nonlinear Program
NLP	–	Nonlinear Program
RLT	–	Reformulation-Linearisation Technique
SLP	–	Successive Linear Programming

Elements and Oxides

Al	–	Aluminium
Al_2O_3	–	Aluminium oxide
C	–	Carbon
$CaCO_3$	–	Limestone
$CaMg(CO_3)_2$	–	Dolomite
CaO	–	Calcium oxide
CO	–	Carbon monoxide
CO_2	–	Carbon dioxide
Cu	–	Copper
Fe	–	Iron
FeMn	–	Ferromanganese
FeO	–	Iron(II) oxide
Fe_2O_3	–	Iron(III) oxide
Fe_3O_4	–	Iron(II, III) oxide
FeSi	–	Ferrosilicon
Mg	–	Magnesium
MgO	–	Magnesium oxide
Mn	–	Manganese
MnO	–	Manganese(II) oxide
MnO_2	–	Manganese dioxide
Mn_2O_3	–	Manganese(III) oxide
Mn_3O_4	–	Manganese(II, III) oxide
P	–	Phosphorus
Si	–	Silicon
SiMn	–	Silicomanganese
SiO_2	–	Silicon dioxide

Introduction (1:2)

- Viktig kapittel: Her skal nysgjerrigheten og interessen til leseren vekkes! Skriv ikke-teknisk.

- 6 nødvendige elementer (ikke nødvendigvis i den rekkefølgen):

1) Centrality of topic. **Motivere hvorfor området er viktig.**
2) Researcher's approach. Sette forskningen i en kontekst – overordnet litteraturhenvisning
3) Motivasjon for forskningen/Rationale. **Hvorfor?**
4) Kort/overordnet problembeskrivelse og forskningsspørsmål: Hensikt/mål (Purpose/Objective/Metode) - Gjerne overordnet (gi beslutningsstøtte til en bedrift med spesifikt problem) og deretter operativt (ny modell, teori, metode). **Hva og Hvordan?**
5) Ha med kort om bidragene/resultater (contributions) **Hva?**
6) Outline/Disposisjon til resten av rapporten.

Introduction (2:2)

- Lengden kan variere for ulike typer prosjekt.
  - Ofte: 1,5 - 4 sider

- Hvis en ikke har bakgrunnskapittel og går rett på litteraturkapittel, bør introduksjonen være lenger for å forstå den valgte litteraturen

- Hvis introduksjonen blir lang (6-10 sider), er det bedre med bakgrunnskapittel

Eksempel

Løyning & Melby, 2018, Masteroppgave «Improving the Robustness of Nurse Schedules in a Real-Life Instance - a Quantitative Analysis Based on Simulation and Rescheduling Under Uncertainty»

Chapter 1

Introduction

Many hospitals, including St. Olavs Hospital in Trondheim, Norway, employ thousands of people with large variety in skills. Many hospital departments are manned round-the-clock, requiring that the employees perform shift work. The personnel schedules are subject to many regulations and preferences originating from both governments, trade unions, managers and employees, which must all be complied with. The result is that making high-quality personnel schedules is a very time-consuming and complex task.

The environment at hospitals is subject to uncertainties in the supply and demand for employees. These uncertainties are difficult to plan for, both in terms of unexpected absence such as sickness leave and a fluctuating number of patients admitted. Consequently, a common problem in health care systems worldwide is shortage of nursing staff combined with the fluctuating nature of patient demand (Lim and Mobasher, 2011). A small staff size leaves the schedules vulnerable to unforeseen events, where the absence of a few employees or unexpectedly high demand may cause a chain reaction of events, where many employees have to work unplanned shifts. The alternative, being understaffed, may negatively affect the health and well-being of both the patients and the employees at work. As making multiple changes to the employees' schedules is both costly to the hospitals and straining for the employees, it is desirable to have some protection against unforeseen events in the hospitals' nurse schedules. This extra protection is what we denote as *robustness*.

The purpose of this thesis is to develop suitable tools to analyze the robustness of nurse schedules, as well as to identify characteristics that improve this robustness. The tools are applied to a real-life case in cooperation with the Department of Neonatal Intensive Care, hereafter also denoted *DNIC*, at St. Olavs Hospital. The purpose is achieved through a four-step process. First, we develop a MIP model that solves the scheduling problem at DNIC. Additionally, multiple model extensions intended to increase the schedule robustness are implemented. Second, we define two models used to simulate the real demand and supply for nurses, based on probability distributions calculated using historical data from DNIC. Third, we develop a MIP rescheduling model, which takes a schedule and an uncertainty realization as input and solves the daily problem of reestablishing the schedule when the real demand exceeds the real supply, using the same rescheduling actions as used in practice at St. Olavs. Fourth and finally, we use multiple measures to assess the robustness of the underlying schedules.

To the best of our knowledge, we are the first to make a rescheduling model that takes simulation results based on probability distributions calculated from real data as input. All models proposed are based on a real-life case, and we believe that our study is among the most realistic ones to date, helping to bridge the gap between theory and practice. Key managerial insights obtained in the thesis include suggestions to DNIC on how the department best can improve the robustness of their schedules, as well as recommendations for the optimal duration of the replanning period used in the rescheduling problem.

The thesis starts with a presentation of useful background information regarding St. Olavs Hospital and DNIC in Chapter 2. The problem is put in a theoretical context in Chapter 3, where we present relevant literature. Chapter 4 contains the problem description, divided into the scheduling problem (Section 4.1), the uncertainty realization (Section 4.2), the rescheduling problem (Section 4.3) and the robustness evaluation (Section 4.4). These are further modelled in the same order in Chapters 5, 6 and 7, respectively. The case description is provided in Chapter 8, and the robustness evaluation is conducted in the computational study in Chapter 9. Finally, concluding remarks and suggestions for future research are presented in Chapters 10 and 11, respectively.

Eksempel

Løyning & Melby, 2018, Masteroppgave *Improving the Robustness of Nurse Schedules in a Real-Life Instance - a Quantitative Analysis Based on Simulation and Rescheduling Under Uncertainty*

Chapter 1

Introduction

Motivasjon for forskningen

Motivasjon for området

Problem-beskrivelse

Many hospitals, including St. Olavs Hospital in Trondheim, Norway, employ thousands of people with large variety in skills. Many hospital departments are manned round-the-clock, requiring that the employees perform shift work. The personnel schedules are subject to many regulations and preferences originating from both governments, trade unions, managers and employees, which must all be complied with. The result is that making high-quality personnel schedules is a very time-consuming and complex task.

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Hensikt, mål og metode

To the best of our knowledge, we are the first to make a rescheduling model that takes simulation results based on probability distributions calculated from real data as input. All models proposed are based on a real-life case, and we believe that our study is among the most realistic ones to date, helping to bridge the gap between theory and practice. Key managerial insights obtained in the thesis include suggestions to DNIC on how the department best can improve the robustness of their schedules, as well as recommendations for the optimal duration of the replanning period used in the rescheduling problem.

Bidrag/ resultater

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Disposisjon

Background (hvis inkludert)

- Beskriv/presenter anvendelsesområdet, bransjen, bedriften/organisasjonen
- Kan inneholde noen fakta med tall
- Relasjon til problembeskrivelsen:
  - Det som er for generelt for problembeskrivelse, kan forklares her
  - Dette kapittelet skal være til hjelp for problembeskrivelsen
- Introduser terminologi og begrepsapparat (nurse scheduling, downtime cost, ….)

Literature review (1:4)

- Har to hovedformål
  - Gi leseren en oversikt over hva litteraturen på området oppgaven omhandler
    - Hva er gjort på samme/lignende problem
    - Hvor er de aktuelle løsningsmetodene benyttet tidligere
  - Posisjonere deres eget arbeid inn i litteraturen
    - Argumentere for at dere gjør noe nytt
    - Argumentere for at valgene dere har tatt mhp modellering og løsningsmetode er fornuftige

- Mange studenter slurver med dette siste og viktige formålet!

Literature review (2:4)

5 C'S IN LITERATURE REVIEW WRITING

1. Cite – Siter tidligere forskning på en ryddig og korrekt måte.
2. Compare – Sammenlign ulike funn, teorier og perspektiver.
3. Contrast – Fremhev ulikheter og motstridende resultater.
4. Critique – Vurder kvaliteten, styrkene og svakhetene ved tidligere studier.
5. Connect – Knytt tidligere forskning til din egen problemstilling og vis hvorfor din studie er relevant.

Literature review (3:4)

- Dekker ofte et bredere område i prosjektoppgaven enn i masteroppgaven.
  - Prosjekt: Først og fremst fokus på problemet dere studerer. Hva er gjort på tilsvarende eller lignende problemer?
  - Master: Her kan fokus gjerne være også på metoder som er brukt på tilsvarende/lignende problemer og/eller som ligner på deres metode
- Unngå opplisting
- I introduksjonen til kapittelet, bør en skrive hvordan en avgrenser litteraturoversikten. Forklar hvordan kapittelet er bygd opp og hvordan dere har gått frem for å finne relevant litteratur

Eksempel

2.2 Literature search strategy

Our search strategy for relevant literature is influenced by Hartmann and Briskorn's (2010) survey article on the variants and extensions of the RCPSP problem. The survey article provides a thorough overview of the key features of this type of problem, and has therefore shaped the structure of this review. To find subsequent relevant literature, searces for RCPSP related terms, such as *RCPSP + MMRCPSP* have been conducted on Google Scholar. All the terms used in conjunction with *RCPSP* in the search are presented in the second row of Table 2.1. The first row of the table is a description of the related groups of search words. The most relevant articles resulting from the searches are accounted for in this review.

Table 2.1: Search words used in conjunction with the term *RCPSP* for literature review.

General terms	Objective function	Activity characteristics	Resource characteristics	Temporal constraints
MMRCPSP MSRCPSP Exact method Solution method Decomposition	Objective fnc Tradeoff Quality	Activity crashing Precedence Location Multi-site Value	Hierarchy Capacity Skill, Multi skill Renewable	Release, Deadline Night, Day Traveltime Transfertime Discrete Continous Setup time

Sørøy & Gullhav, 2025, *An integrated project and personnel scheduling problem with resource transportation and preemption*

Eksempel

Litteraturen sammenliknes (og kontrasteres) langs flere dimensjoner: ett eller prosjekter, en eller flere modus, homogen/heterogent personell, skift, preemption (avbrytbare aktiviteter). Kunnskapshull presenteres.

As is the case in the RCPSP, several papers consider the single-project (e.g., see Alfares & Bailey, 1997; Dhib et al., 2015; Maenhout & Vanhoucke, 2016; Van Den Eeckhout et al., 2019) and multi-project cases (e.g., see Dodin et al., 1998; Felberbauer et al., 2019; Heimerl & Kolisch, 2010; Kolisch & Heimerl, 2012). Furthermore, many of the papers include activities in their problems. In Heimerl & Kolisch (2010), however, they present a multi-project problem but do not include activities in the mathematical formulation. In this way, they do not have to consider the precedence relations within each of the projects. In Kolisch & Heimerl (2012), on the other hand, they include activities in their model formulation.

Another dimension to consider is whether the projects and/or the activities are multi-modal. Alfares & Bailey (1997), Maenhout & Vanhoucke (2017), and Volland et al. (2017) include one mode only in their single-project problems, while Van Den Eeckhout et al. (2019) extend the problem to support multiple modes. In the multi-project problem from Heimerl & Kolisch (2010), there is only one mode for each project.

As in the PSP case, papers that consider the integrated project and personnel scheduling problem either assume a homogeneous or heterogeneous workforce. Alfares & Bailey (1997), Van Den Eeckhout et al. (2019), and Volland et al. (2017) assume that all the employees in the problem possess the same skill set. Consequently, all the employees are able to perform all the activities that are to be scheduled. On the other hand, Bellenguez-Morineau & Néron (2007), Cordeau et al. (2010), and Heimerl & Kolisch (2010) make a more general assumption by allowing for heterogeneous skill sets. If the workforce is homogeneous, deciding which specific employee to work where and when is not necessarily required. For example, both Alfares & Bailey (1997) and Van Den Eeckhout et al. (2019) only decide how many employees should work in each schedule. For the heterogeneous case, Bellenguez-Morineau & Néron (2007), Cordeau et al. (2010), and Heimerl & Kolisch (2010) all plan where and when each of the employees should work.

Most of the consulted papers that consider the IPPSP use some sort of shifts in their mathematical formulation (e.g., see Dodin et al., 1998; Drezet & Billaut, 2008; Fırat & Hurkens, 2012; Heimerl & Kolisch, 2010; Kolisch & Heimerl, 2012; Volland et al., 2017). Other papers such as Alfares & Bailey (1997), Maenhout & Vanhoucke (2016, 2017), and Van Den Eeckhout et al. (2019) aggregate the shifts into structures that are similar to schedules and stints. In Van Den Eeckhout et al. (2019), they provide the constraints used to generate the schedules in their problem as well. In their model, they use time periods similar to shifts.

Based on the presented papers above, we can see that preemption has not been regarded considerably for the IPPSP. However, Felberbauer et al. (2016) and Felberbauer et al. (2019) include preemption on the project level, which means that each project essentially includes activities that need to be fulfilled in sequential order. This is in contrast to the integer-preemptive RCPSP from Demeulemeester & Herroelen (1996), in which the activities themselves can be preempted one or more times during the execution. Dhib et al. (2015) consider a preemptive project staffing and scheduling problem in which some of the activities can be preempted in discrete time steps, and they do not consider setup time nor costs if the activities are preempted.

An overview of the 15 presented papers for the IPPSP is shown in Table 1, which compares the features of the IPPSPs in the respective papers with the variant considered in our paper. Here, we can see that most of the papers include multi-skilled personnel and allow for external personnel, while only one paper considers the multi-modal case. A few papers include preemptive activities. However, none of these papers consider transportation of resources.

Eksempel

- Bruk gjerne tabeller for å systematisere, oppsummere og sammenlikne med egen oppgave (connect):

Sørøy & Gullhav, 2025, *An integrated project and personnel scheduling problem with resource transportation and preemption*

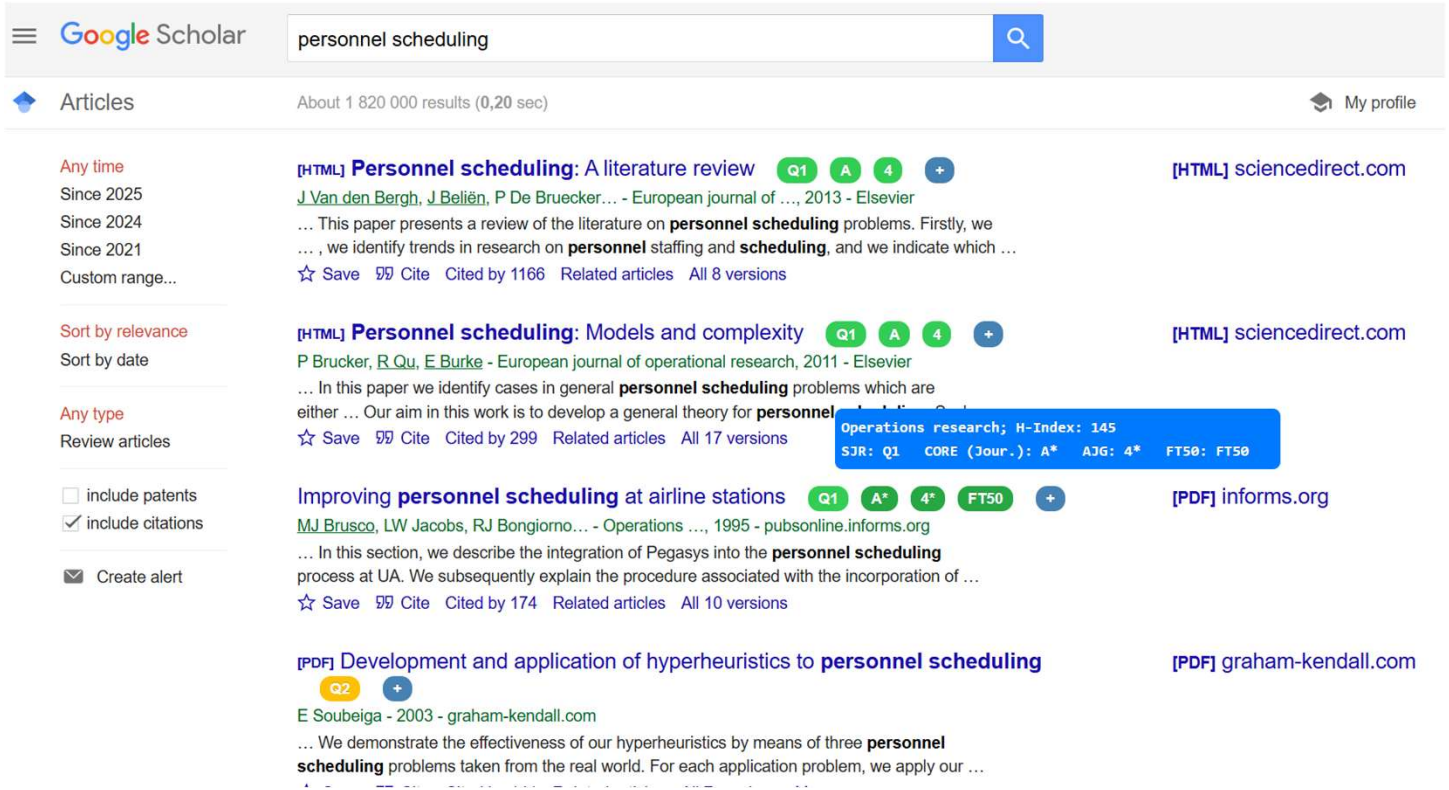
Table 1 Summary of the presented literature of the IPPSP. The abbreviations are as follows: Obj=Objective Pro=Number of projects, S=Skills, MM=Multi-modal, Pre=Preemption, RT=Resource transportation, ER=External resources

Paper	Obj	Pro	S	MM	Pre	RT	ER
Alfares & Bailey (1997)	Min financial costs	One					
Bellenguez-Morineau & Néron (2007)	Min makespan	One	✓				
Cordeau et al. (2010)	Min makespan	One	✓				✓
Dhib et al. (2015)	Min makespan	One	✓		✓		
Dodin et al. (1998)	Min financial costs	Several	✓				
Drezet & Billaut (2008)	Min maximal lateness	One	✓				
Felberbauer et al. (2016)	Min staff costs	Several	✓		✓		✓
Felberbauer et al. (2019)	Min external costs	Several	✓		✓		✓
Fırat & Hurkens (2012)	Min makespan	One	✓				✓
Heimerl & Kolisch (2010)	Min staff costs	Several	✓				✓
Kolisch & Heimerl (2012)	Min staff costs	Several	✓				✓
Maenhout & Vanhoucke (2016)	Min staff costs	One					✓
Maenhout & Vanhoucke (2017)	Min staff costs	One					✓
Van Den Eeckhout et al. (2019)	Min staff costs	One		✓			✓
Volland et al. (2017)	Min staff costs	One					✓
This paper	Min resource costs	Several	✓	✓	✓	✓	✓

Literature review (4:4) – kilder

- En mye brukt kilde: https://scholar.google.com/
  - Men du må passe på at artiklene er publisert i anerkjente fagfellevurderte tidsskrifter, eller at de er skrevet av anerkjente forfattere innen feltet.
  - Merk at det (på venstre side) finnes et «Avansert søk» hvor du kan gjøre et mer spesifisert søk.
- Andre databaser:
  - https://www.scopus.com/
  - https://www.ntnu.no/ub (NTNU Universitetsbiblioteket)
- Noen forfattere laster opp artiklene sine før de blir publisert, spesielt når de er under vurdering (review). Du kan finne slikt materiale på: https://arxiv.org/, https://www.ssrn.com/, https://www.researchgate.net/
  - Men vær forsiktig! Disse artiklene er nødvendig ikke fagfellevurderte.
- Se også mer informasjon på Innsida: https://i.ntnu.no/wiki/-/wiki/Norsk/Avanserte+litteraturs%C3%B8k
- En veldig god måte å finne gode artikler på, er å se på referansene i de gode artiklene du allerede har funnet!
  - Google scholar finner også artikler som har sitert en gitt artikkel.

Verktøy for å sjekke tidsskriftkvalitet “Rapid Journal Quality Check”



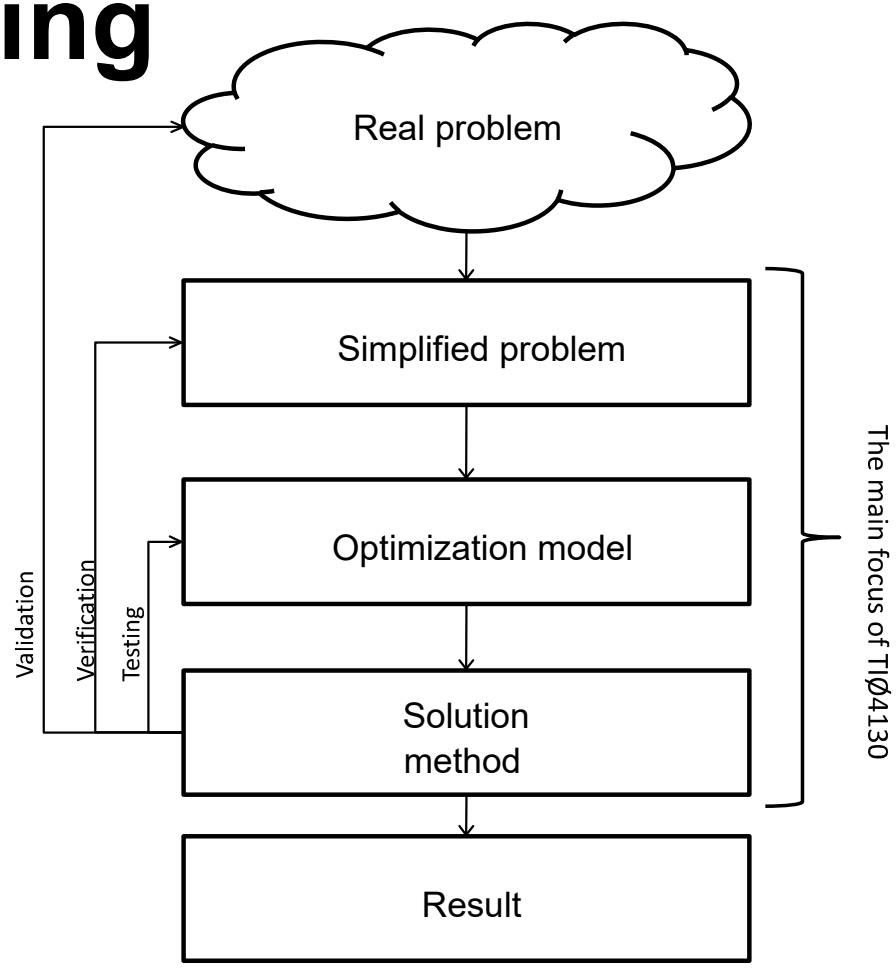
Anerkjente tidsskrifter

- Det eksisterer forskjellige rangeringssystemer for tidsskrifter, blant annet:
  - Impact factor
  - Register over vitenskapelige publiseringskanaler (Norsk rangeringssystem: 2 er best)
  - UK ABS: https://charteredabs.org/academic-journal-guide/academic-journal-guide-2024
- Topprangerte operasjonsanalyse:
  - Operations Research
  - Management Science
  - Mathematical Programming
  - European Journal of Operational Research (EJOR)
  - Naval Research Logistics
  - IIE Transactions
  - INFORMS Journal on Computing
  - Computers and Operations Research (C&OR)
  - Transportation Science
  - Production and Operations Management (POM)
  - Journal of Scheduling
- Diskuter også med veileder

Matematisk modellering

- *Simplified problem* er en ikke-matematisk beskrivelse av (ressursplanleggings)problemet man studerer.
  - Danner grunnlaget for den matematiske beskrivelsen i *optimization model*
- *Simplified problem* og *Optimization model* (Mathematical model) presenteres oftest i ulike kapitler.
  - *Simplified problem* benevnes ofte «Problem description» i rapporten

Figur fra undervisningen i TIØ4130



Problem description (1:3)

Se for deg følgende situasjon:

Du har fem minutter til å forklare problemet til noen som ikke kjenner det fra før. Etter disse fem minuttene skal de ha nok kunnskap til å skjønne:

- Hva problemet og «settingen» er (sistnevnte kan evt. være i et bakgrunnskapittel),
- hvilken informasjon som er tilgjengelig (*input*),
- hva formålet (*objective*) er,
- hva *begrensingene* er,
- hvilke *beslutninger* som skal tas, og evt.
- Hvordan ser en løsning (plan) ut - *output*

Dette skal helst beskrives detaljert nok til at vedkommende kan sette opp en matematisk modell av problemet.

Problem description (2:3)

- Oppfattes ofte som vanskelig
- Beskriv problemet kort og konsist. Vær konkret. Normalt ikke mer enn 1-2 sider (unntak kan forekomme)
- Bør generelt ikke inneholde data (størrelsesorden kan evt. indikeres)
- Skill på problemantakelser og modelleringsantakelser
  - Ikke ha med modelleringsantakelser og modelleringsvalg her, mens antakelser rundt selve problemdefinisjonen må være med
- Basert på problembeskrivelsen, skal en kunne sette opp modell. Sjekk om alt er med (målfunksjon,….)..

Problem description (3:3)

Rekkefølge:

- Overordnet formål
- Beskriv entitetene som forekommer i problemet
  - Hvilke egenskaper og «input data» har hver entitet
  - Beskriv en og en entitet med tilhørende egenskaper
  - Beskriv relasjoner til tidligere introduserte entiteter
    - Hvordan «samhandler» entitetene
    - Hvilke egenskaper og «input data» kan knyttes til relasjonen/samhandlingen?
    - Merk, at avhengighetene mellom entitetene avgjør rekkefølgen de introduseres i
- Beskriv hvilke beslutninger som skal tas i deres problem
  - Hva kan beslutningstakeren faktisk endre på
  - Merk at hva som er input og hva som er beslutning avhenger av beslutningsnivå
- Beskriv hvordan man evaluerer kvaliteten på et sett av beslutninger
  - Dvs. beskriv objektivfunksjonen deres med ord

Eksempel

3 Problem Description

This chapter presents the problem studied in this thesis and outlines the roles of dairy facilities, routes, and vehicles. Additionally, it introduces the requirements and limitations that must be considered in the planning process.

Norsk Melkeråvare has the responsibility to deliver milk to a set of dairy facilities geographically dispersed throughout Norway. Each of the dairy facilities produces products within one or two product categories. Each week, each dairy facility places an order specifying daily volume requirements for the following week. This ordered volume is henceforth referred to as the dairy facility's demand and must be met. The orders can be for multiple types of milk. Some dairy facilities order multiple types of milk, while others only order one type. Dairy facilities producing milk powder and regulation cheese serve as regulatory facilities, but can also place orders that must be met. As milk is a perishable product, it has to be used in production within a given number of hours after it is delivered to a dairy facility, depending on the type of products produced at the dairy facility.

Norsk Melkeråvare collects milk from farms and deliver it to dairy facilities to ensure that the dairy facilities' demand is met. This collection process is done using a predefined set of routes. Each route has a length and duration, as well as an endpoint. From the endpoint of each route, the milk is delivered to one or more dairy facilities. It is assumed that each route has an initial inventory and produces consistent daily volumes of milk, which may include multiple types. Each farm has storage tanks that has to be cleaned within a limited number of hours to prevent high bacteria levels in the tank. For the farms, it is assumed that the maximum number of hours milk can be stored sufficiently limits the volume such that storage capacity can be neglected.

The milk is collected using a set of vehicles. Each vehicle has a designated dairy facility where it starts and ends every day and an average cost per kilometer. The time it takes to pump milk on and off a vehicle varies with the volume of milk collected, and whether the vehicle has a trailer or not. Due to hygienic requirements, each vehicle has to be cleaned at the end of the day. The cleaning requirements also limits the number of hours milk can be stored on a vehicle. Furthermore, a vehicle has a maximum number of hours it can drive each day, including pumping time, driving time and cleaning time. A vehicle can operate on multiple routes per day, as long as its total driving time remain within the maximum allowed limit. Vehicle size limits access to certain roads, bridges, dairy facilities, and farms, making some routes and dairy facilities incompatible with larger vehicles.

The decisions to be made are on which days each route should be driven, what dairy facilities the collected milk should be delivered to and which vehicle to use. Due to the short time period milk may be in inventory on farms and dairy facilities, no inventory holding costs are considered. The goal is to minimize the costs related to milk collection with regards to driving costs per kilometer and costs for driving outside regular working hours, while meeting the demand of the dairy facilities and collecting all milk. The model assumes instant driver shift changes. Hence, overtime is not considered. To account for government subsidies, parts of the distance between a route and its assigned dairy facility is subsidized and thus subtracted from the cost calculation. If milk is delivered to a dairy facility that is closer than the assigned facility, the resulting cost is floored at zero to avoid negative values. If the milk is instead delivered to a facility located farther away, the subsidy still applies: parts of the cost corresponding to delivery to the assigned facility is covered.

Eksempel: entities

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Eksempel - relations

3 Problem Description

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Eksempel - decisions

3 Problem Description

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Eksempel - evaluering

3 Problem Description

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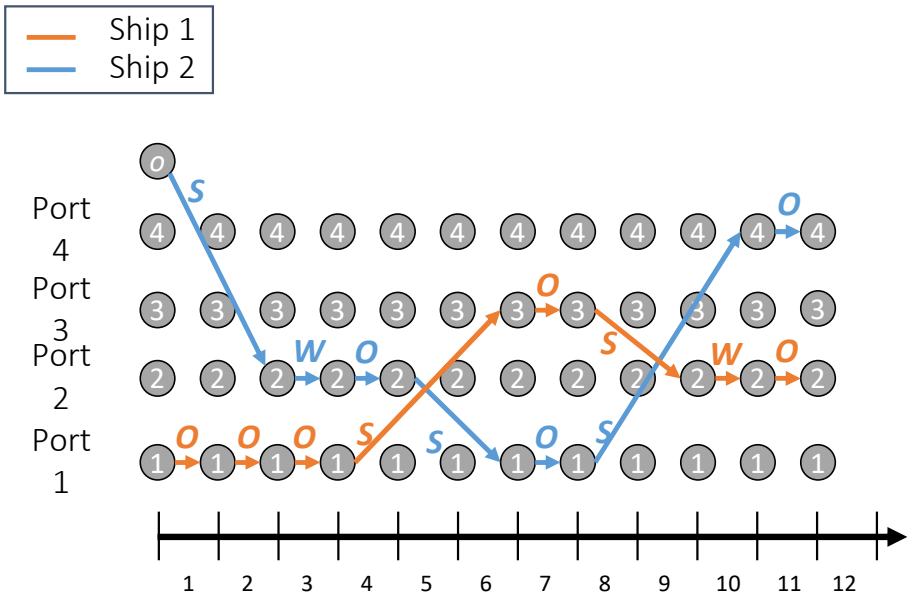
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Mathematical model - Introduction

- Deles normalt opp i flere underkapitler:
  - antakelser (modeling assumptions)
  - evt. modelleringstilnærming (eks. bruk av diskret tid eller 2-stegs stokastisk programmering)
  - notasjon
  - modell
  - alternativ/utvidet/forbedret modell (kanskje aktuelt)
    - F.eks. gyldige ulikheter, symmetribrytende restriksjoner som en kanskje ønsker spesifikt å teste effekten av i beregningsstudien
    - F.eks. path flow-modell dersom en allerede har presentert en arc flow-modell

Mathematical model - Assumptions

- Vær tydelig på alle modelleringsantakelser og valg (discrete/continuous time, end of time horizon effects,…)
- Bruk gjerne figurer



Mathematical model - Notation

- Kan skrives i tekst eller med tabeller (Indices, Sets, Parameters, Variables), eller evt. begge deler. Hvis stor modell, anbefales tabeller for å oppsummere notasjon.
- Indekser og variable med små bokstaver
- Parametere: store. Mengder: store og gjerne \mathcal i Latex
- Vær ekstremt nøye med å deklarere alt en bruker i modellen, og ikke deklarer symboler som du ikke bruker
- Bruk gjerne notasjon som er vanlig i litteraturen
- Begrens bruk av greske bokstaver, hvis dette ikke er vanlig
- Bruk flere bokstaver for mengder, parametere og variable hvis behov (men da bruk superscript), men bruk gjerne kortere navn enn i Python/Gurobi.

$$min\ z = \sum_{t \in \mathcal{T}} \sum_{m \in \mathcal{M}_v} \sum_{j \in \mathcal{N}^T} \sum_{i \in \mathcal{N}^T} \sum_{v \in \mathcal{V}} C_v^S T_{vijt}^{SD} x_{vijmt} + \sum_{t \in \mathcal{T}^{DAY}} \sum_{v \in \mathcal{V}} C_v^F \delta_{vt}$$

Sets

$\mathcal{A}$ - Set of market areas

$\mathcal{I}$ - Set of all ATM locations, both existing and possible new locations

$\mathcal{I}_a^A$ - Set of all ATM locations that belongs to market area a ($\mathcal{I}_a^A \subset \mathcal{I}$)

$\mathcal{I}^E$ - Set of locations with existing ATMs in the beginning of the planning horizon ($\mathcal{I}^E \subset \mathcal{I}$)

$\mathcal{I}^N$ - Set of new ATM locations without an exisiting ATM, ($\mathcal{I}^N \subset \mathcal{I}$) and ($\mathcal{I}^N = \mathcal{I} \setminus \mathcal{I}^E$)

$\mathcal{I}_{mk}^C$ - Set of ATM locations that are within a distance D_k^C of demand node m

$\mathcal{M}$ - Set of demand nodes

$\mathcal{T}$ - Set of time periods

$\mathcal{K}$ - Set of covering criterias

Revenue and cost parameters

R_{it} - Revenue generated by the ATM on location i in period t

C_{it}^{TO} - Total cost for running an ATM at location i in time period t, excluding depreciation costs

C_t^{TDN} - Cost of new ATM given installation in time period t

C_t^T - Fixed cost of terminating an ATM in time period t

C_t^R - Fixed cost of relocating an ATM in time period t

C_t^I - Fixed cost of installing a new ATM in time period t

C_t^S - Storage cost per stored ATM in time period t

Covering parameters

$\underline{I}_{at}^A$ - Minimum number of open ATMs in market area a in time period t

D_k^C - Maximal distance from demand node m to an ATM in covering criteria k

N_m^C - Number of customers in demand node m

P_{tk}^C - Minimum covered proportion according to covering criteria k in time period t

Time parameters

T_i^{ATM} - Time in service of the ATM at location i in the beginning of the planning horizon

T_i^N - Planned time period for replacing the ATM at location i ($i \in \mathcal{I}^E$)

$\overline{T}^R$ - Maximum service time for an ATM for it to be allowed to relocate

$\overline{T}^{SL}$ - Maximum service life of an ATM

$\overline{T}^S$ - Maximum storage time

Mathematical model – Basic model presentation

- Alternative presentasjonsformer:
  - Kan skrives samlet og alle restriksjonene beskrives under.
  - Kan skrive ulike deler (inventory management, routing, allocation) og deretter forklare restriksjonene tilhørende hver del etter modelldelen.
  - Kan presentere en og en restriksjon med tilhørende forklaring.
- Valget avhenger av kompleksiteten til restriksjonene og størrelsen på modellen.

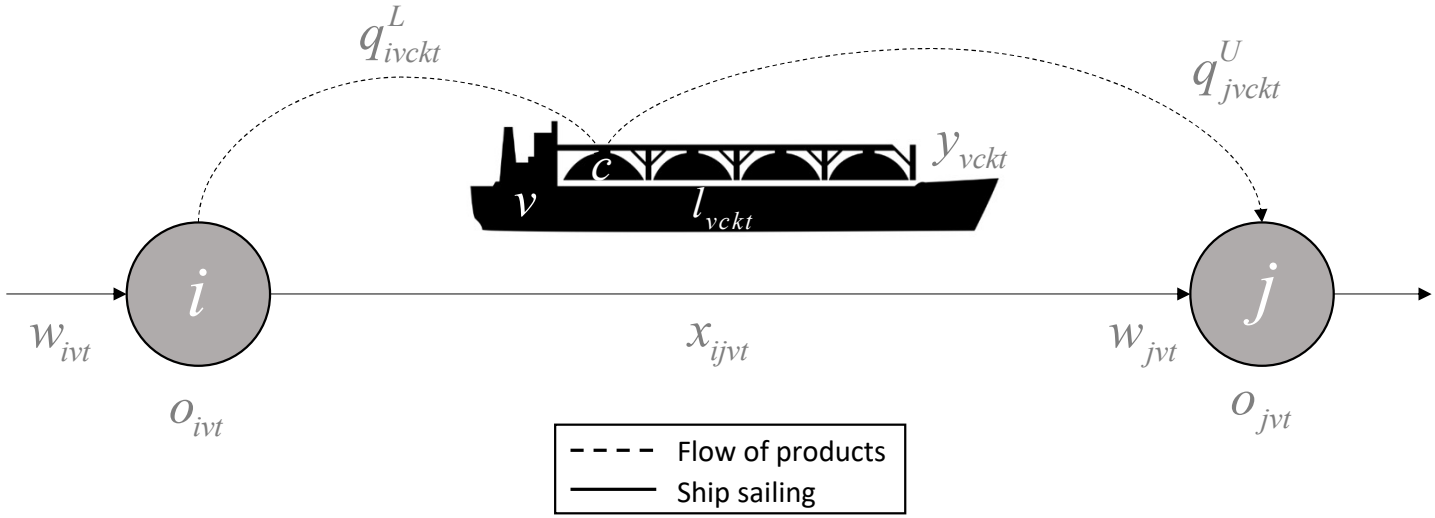
- Bruk gjerne appendix for full modell

Mathematical model - utvidelser

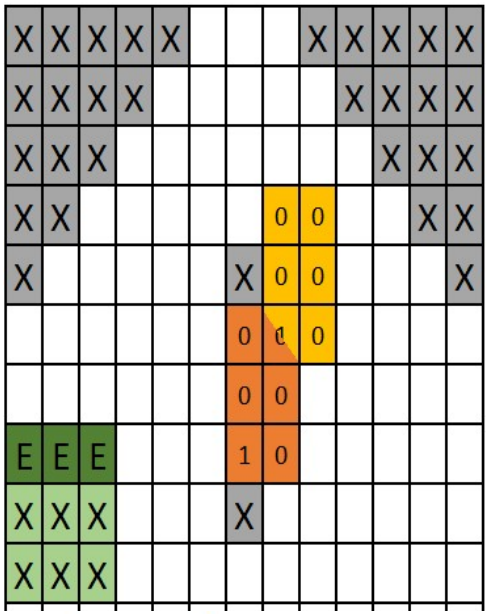
- Alternative modeller – for sammenligning
- Strengthening the model
  - Valid inequalities
  - Symmetry breaking constraints
  - Extended formulation
- Extended model – inneholder en rikere modell (færre antakelser)
- Alle disse kan være et delkapittel under «Mathematical model» eller et eget kapittel. Dette avhenger av omfanget.

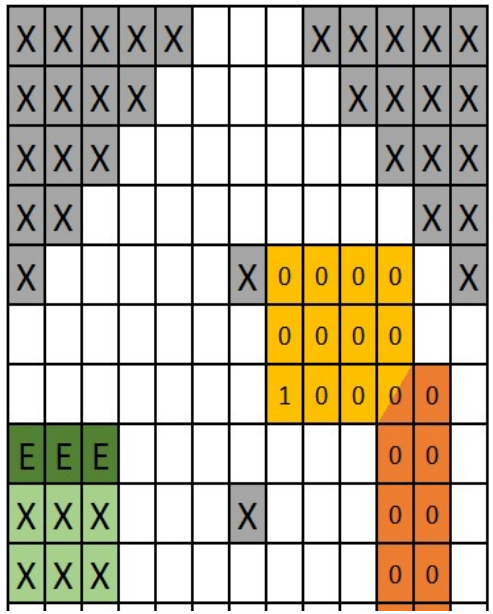
Mathematical model - presentasjonsråd

- Bruk figurer hvis det er vanskelig å forklare eller et uoversiktlig problem.
- **Figur til forklaring av variabler:**



Figur til forklaring av restriksjoner





$$\sum_{i'=i}^{i+S_c^L-1} \sum_{j'=j}^{j+S_c^W-1} x_{i'j'c} \leq 1, \qquad c \in \mathcal{C}, i \in \mathcal{I}_c, j \in \mathcal{J}_c \quad (4.7)$$

$$\sum_{i'=max(i-S_{c'}^L+1,1)}^{min(i+S_c^L-1,|\mathcal{I}_{c'}|)} \sum_{j'=max(j-S_{c'}^W+1,1)}^{min(j+S_c^W-1,|\mathcal{J}_{c'}|)} x_{i'j'c'} \leq M_{cc'}(1-x_{ijc}),$$

$$c \in \mathcal{C}, c' \in \mathcal{C}\backslash\{c\}, i \in \mathcal{I}_c, j \in \mathcal{J}_c \quad (4.8)$$

Mathematical model - Formuleringsråd

- Eneste tall som aksepteres er logiske tall (normalt 0 og 1)
- Stor M: Beskriv verdien som den bør ha for å være så stram som mulig (evt. i eget underkapittel)
- Lesbarhet: vurder å først ha ulineær, så lineær (evt. i eget underkapittel):

$$\sum_{j\in\mathcal{I}}\sum_{p=t}^{|\mathcal{T}|} w_{ij(t+1)p}\theta_{it} \le \overline{T}^R, \qquad i\in\mathcal{I},\ t\in\mathcal{T}\mid t<|\mathcal{T}|. \tag{6.21}$$

Linearization of constraints (6.21)

$$\theta_{i(t-1)} - (\overline{T}^{SL} - \overline{T}^R)\sum_{j\in\mathcal{I}}\sum_{p\in\mathcal{T}}(1-w_{ijtp}) \le \overline{T}^R, \qquad i\in\mathcal{I}^E, t\in\mathcal{T}|t\ge T_i^N \tag{6.27}$$

$$\theta_{i(t-1)} - (\overline{T}^{SL} - \overline{T}^R)\sum_{j\in\mathcal{I}}\sum_{p\in\mathcal{T}}(1-w_{ijtp}) \le \overline{T}^R, \qquad i\in\mathcal{I}^N, t\in\mathcal{T}|t>1 \tag{6.28}$$

Constraints (6.21) are easily linearized with no increase in the model size. Further, by using x_{it} instead of the scalar 1 and by choosing the values of the Big M's wisely, the model size can actually be reduced by includning constraints (6.19) and (6.20) in constraints (6.27) and (6.28).

Mathematical model – Typisk feil

- Sjekk at hver indeks i restriksjonen enten blir summert over eller er gjort for alle restriksjoner:

$$\sum_{\tau=t}^{t+(H-1)} \sum_{m \in \mathcal{M}_v} \sum_{i \in \mathcal{N} \cup \{0\}} (y_{vim\tau} + x_{vijm\tau}) - H\delta_{vt} \leq 0\,, \qquad v \in \mathcal{V}, t \in \mathcal{T}^{DAY}$$



- Sensorene har irriterende gode briller

Solution method

- Mer vanlig å inkludere i master- enn i prosjektrapport
- Kan ha veldig ulik form avhengig av metode (eksakt/heuristikk)
- Forklar løsningsmetode på minst to nivå
  - Først overordnet. Bruk gjerne flytskjema uten for mye tekst
  - Forklar hver del av løsningsmetoden i detalj. Bruk gjerne pseudokode (abstraher, ikke for detaljert, sjekk at alt stemmer og er fullstendig)
- Metodevalg:
  - Motiver for valg av metode innledningsvis i kapitlet (og ikke bare «fordi veileder foreslo det…»)
  - Kan gjerne referere til litteratur som har brukt tilsvarende metode med suksess (innledningsvis og underveis for detaljvalg som gjøres)

Solution method - flytskjema

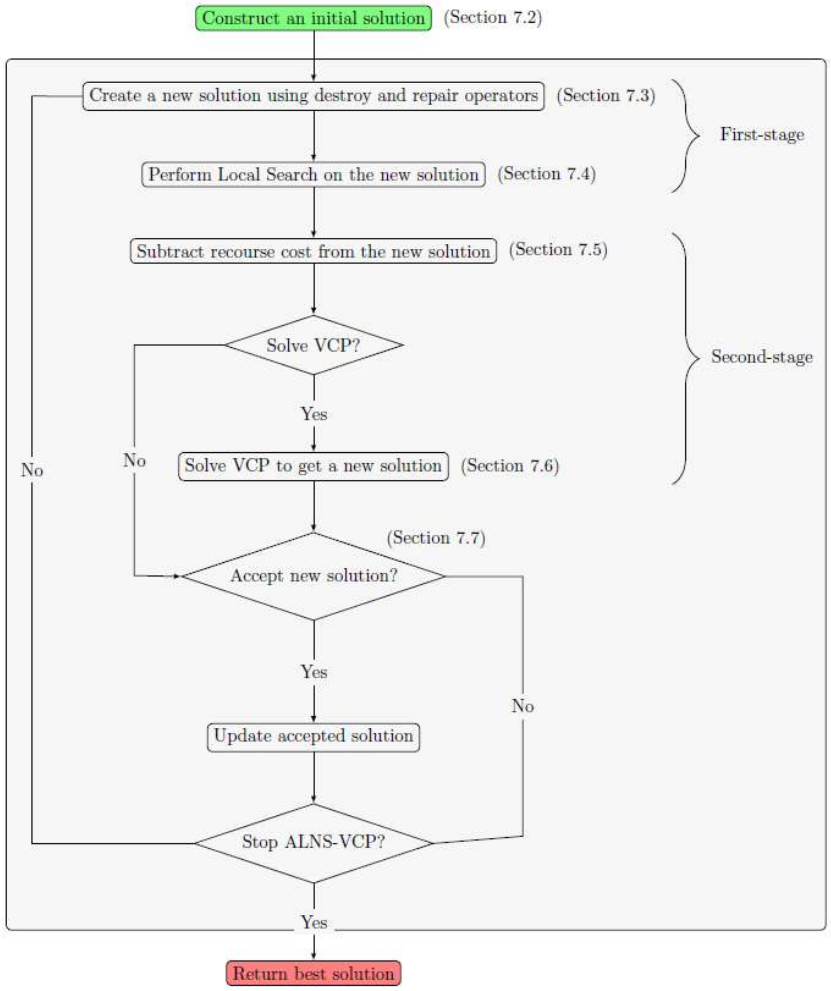


Figure 7.1: Flow chart representing the different components of the iterative ALNS-VCP matheuristic.

Pseudokode

Algorithm 1 Overview of Matheuristic

Calculate minimum number of visits (p_i), compartment visits (p_{ik}), operation periods (σ_i) and time periods for mandatory visits (z^1_{jk}, z^2_{jk}, z^3_{jk})
initialRoutes $\leftarrow$ *RouteGenerator*(p_i,p_{ik},σ_i,z^1_{jk}, z^2_{jk}, z^3_{jk})
fix sequence of initialRoutes $\rightarrow$ *Sequence*
continue := true
while continue **do**
  CurrentSolution = *InventoryProblem(Sequence)*
  Save inventory limit violations from InventoryProblem(Sequence) $\rightarrow$ *inventoryViolations*
  if inventoryViolations is_empty **then**
    continue := false
    return CurrentSolution and CurrentObjectiveValue
  end if
  Routes = *RouteGenerator(inventoryViolations)*
  Fixed sequence of Routes $\rightarrow$ *Sequence*
end while
Run *Improvement heuristic* with CurrentSolution and CurrentObjectiveValue

Implementation

- Dette kapittelet trengs bare om det er behov for å bruke mye plass på å forklare hvordan de ulike metodedelene er implementert og sammenhengen mellom dem.
- Aktuelt hvis bruk av mange ulike programpakker: Matlab, Excel, Gurobi, Python,…..
- Aktuelt hvis man har laget egen instansgenerator å beskrive denne i et slikt kapittel
- Hvis det er kort, kan det inngå i innledningen til Computational study (vanligst)

- Kan være like aktuelt å ha et eget kapittel på «Case study and test instances»

Computational study

- Krevende. Skrives ofte under tidspress.
- Her diskuteres bare kort hvordan det skal skrives, mens innholdet diskuteres i en senere forelesning
- Tenk på hva dere ønsker å formidle
- Typisk oppbygning:
  - Innledning
  - Instansinformasjon (hvis ikke i eget kapittel)
  - «Teknisk» studie – fokus på løsningstider og løsningskvalitet (gap)
  - «Økonomisk» studie – fokus på «managerial insights»

Innledning til:

Computational Study

The purpose of this chapter is to present an overview of the results from the testing of the MIRP-UC with both an exact solution method and the matheuristic solution methods developed in Chapter 5. When testing the model formulation with the exact solution method and the valid inequalities proposed in Section 4.3, three test cases of different sizes are utilized. Further, when testing the matheuristic methods, additional test cases are used to further explore the potential of the alternative solution approach. This chapter is organized into two main section. First, in Section 6.1, we present the results utilizing the exact solution method as well as the effectiveness of the proposed valid inequalities. Following is Section 6.2, presenting the results of the matheuristic solution method.

All test instances of our mathematical programming models are solved using Mosel Xpress-MP. Mosel Xpress-MP is run on a Hewlett Packard 64-bit Windows 7 Enterprise PC with Intel(R) Core(TM) i7-3770 3.40 GHz processor and 16,0 GB (15.9 GB usable) RAM.

- Hensikt
- Rekapitulere forskningsspørsmål og mål med oppgaven
- Oppbygning av kapittelet
- Presenter programmerings-språk, software og hardware (her eller under Implementation)

Concluding remarks

- Skal være kort (1-2 sider)
- Skal være forskjellig fra Abstract/Summary

- Forklar meget kort om problem og løsningsmetoder.
- Beskriv hovedbidrag
- Beskriv hovedresultater
- Si noe om betydningen av resultatene
  - Forklar mer detaljert om hvordan du har svar forskningsspørsmålene som beskrevet under «purpose». Hvor nær målet kom du? Hva lærte vi?

Future research

- Viktigere i prosjekt- enn masterrapporten

- Diskuter ulike dimensjoner
  - Videreutvikle modell og løsningsmetode
  - Ta inn nye elementer i problemet (f.eks. rikere/mer detaljer, utvide, flere mål, inkluder usikkerhet)

Eksempel på struktur (revisited)

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References

- Ulike stiler. **Vær konsistent!** Bruk gjerne Bibtex-referanser hentet fra Google Scholar hvis du skriver i LaTeX.
- Mangelfull referering trekker ned inntrykket
- Hvert tidskrift setter opp sine retningslinjer (Transportation Science):
- **Article in a journal:**
  Flynn J, Gartska SK (1990) A dynamic inventory model with periodic auditing. *Oper. Res.* 38(6):1089-1103.
- **Book**
  Makridakis S., Wheelwright SC, McGee VE (1983) *Forecasting Methods and Applications,* 2nd ed. (John Wiley & Sons, New York).

References

- **Edited Book**

Martello S., Toth P. (1979) The 0-1 knapsack problem. Christofides N, Mingozzi A, Sandi C, eds. *Combinatorial Optimization* (John Wiley & Sons, New York), 237-279.

- **Conference Proceedings, Published**

Hill RR (1996) An analytical comparison of optimization problem generation methodologies. Medeiros DJ, Watson EF, Carson JS, eds. *Proc. 1998 Winter Simulation Conf.* (Institute of Electrical and Electronics Engineers, Washington DC), 609-615.

- **Presentations**

Haltiwanger J (2004) What do we know (and not know) about the contribution of young businesses to economic growth? Presentation. Entrepreneurship in the 21st Century Conference, March 26, Office of Advocacy, U.S. Small Business Administration, Washington, DC.

- **Dissertations, Unpublished**

Yi Y (1998) Developing and validating an observational learning model of computer software training. Unpublished doctoral dissertation, University of Maryland, College Park.

Referering i tekst

- En eller to forfattere:
  - Fagerholt (2001) describes a problem…
  - Andersson and Stålhane (2014) present a model for….
- Tre eller flere forfattere
  - Andersson et al. (2012) introduce the use of…

- In Andersson and Tomasgard (2010), a stochastic planning problem is….. (her refereres det til artikkelen og ikke forfatterne.)

- IRPs are known to be even harder to solve (Bertazzi and Speranza, 2012).

- Best med forfattere og årstall for referanse. [1] brukes ofte i artikler hvor det er viktig å spare plass (anbefales ikke).

- Hvis du er i tvil om du skal referere – referer! (unngå mistanke om plagiat)

Appendices og håndtering av kode

- Appendix:
  - Appendix A (A.1, A.2…), Appendix B, …..
  - Full notasjon og ulike fulle modeller kan gjerne inngå i appendiks
  - Mer beregningsresultater (?)
  - Inkluder ikke kode

- Kode- og datafiler kan evt. lastes opp i Inspera, men bør i alle fall sendes til veileder. Kodefiler må være kommentert. Sensor kan også be om å se dette.

- Masteroppgave leveres i Inspera. Alt som lastes opp der er tilgjengelig for alle. Unngå derfor datafiler etc. som kan være konfidensielle.

Språk og tekstbehandingssystem

- Engelsk/Norsk
- LaTeX/Word

Generelle råd - skrivingsrekkefølge

- Typisk rekkefølge på skrivingen (avhenger noe av type prosjekt):
  - Bakgrunn (om det skal være med)
  - Problembeskrivelse
  - Litteratur med referanser (en må i alle fall starte tidlig med å lese og strukturere litteraturen, skrivingen kan en evt. vente noe med)
  - Modell
  - Løsningsmetode
  - Beregningsresultater
  - Forord, oppsummering, introduksjon, konklusjon, future research

Generelle råd

- Referer til prosjektrapporten som «project report» og masteroppgaven som «master thesis»
- Tenk på at dere skal hjelpe leseren med å forstå innholdet – vær så pedagogisk som mulig
- Det er ikke mer imponerende med en lang rapport!
  - Det tar tid å skrive kort, men blir ofte bedre og mer gjennomarbeidet
  - Sørg for å få med beskrivelser og forklaringer slik at leseren forstår detaljene
  - Ikke gjenta dere selv for ofte og vær presis (men om en kort gjentakelse kan påminne leseren om noe vanskelig han/hun kanskje har glemt, kan det likevel være gunstig)
  - Bruk plass på figurer, flytskjema, tabeller…..
  - Vær klar over at en har mer plass enn i en artikkel
  - Sjekk for grammatikk- og andre skrivefeil – er veldig irriterende og unødvendig med for mange slike!
- Nummerer sidene
- Unngå gjerne fotnoter

Generelle råd - Delkapitler

- Bruk delkapitler for å bedre struktur.
  - Chapter 4.
  - Section 4.1
  - Subsection/Section 4.1.1

- Hvis et kapittel inneholder flere delkapitler, introduser delkapitlene innledningsvis i hovedkapitlet

> Section 6.1 explains the specific applications of the model developed, whereas some general model assumptions are stated in Section 6.2. Section 6.3 gives a step by step explanation of the general model formulation, before a more problem specific simplification is given in Section 6.4. The multi-objective formulation of the objective function is given in Section 6.5.

Språkråd

- I vitenskapelige litteratur brukes mer passiv, men fortsatt prøv å skrive aktivt
- Bruk gjerne nåtid. (Fagerholt (2004) finds that….) – uansett: vær konsekvent!
- Vær forsiktig med **will** (futurum). In this paper, we present….
- Varier språk: Constraints (5.13) state/ensure/enforce/connect…..
- Men ha klart definerte ord for begrep og ikke varier her:
  - sail/move
  - product/commodity
  - employee/worker
- Vær presis og tydelig. Bruk gjerne korte setninger.
- Forkortelser skal alltid introduseres første gang de brukes (og ellers bare i figurer og tabeller). Unngå for mange forkortelser.

Tabell- og Figurråd

- Tabelltekst: Over tabell. Figurtekst: Under figur
- Ha en beskrivende tekst til tabell og figur

- Figurer:
  - Skal hjelpe å forstå teksten
  - Må være nøyaktig og se oversiktlig ut
  - Akser må navngis/benevnes

- Tabeller:
  - Må ikke inneholde for mye innhold
  - Tall må oppgis med passende og konsistent presisjonsnivå
  - Passende bruk av bold og italic fonts.
  - Plasser de beste resultatene (dine) lengst til høyre.
  - Bruk tidsenheter (“seconds” er bedre enn “time”)
  - Inkluder gjennomsnitter

Evalueringskriterier prosjekt- og masteroppgave

	Delpunkt	E/I ^	Maks Poeng	Evaluation	^ E: ekstern sensor / I: intern (veileder)	English
Innledning & teori	Faglig forankring	E	5	Introduction & theory	Er det teoretiske og faglige grunnlaget godt beskrevet slik at arbeidet blir satt inn i fagfeltets internasjonale forskning?	Disciplinary foundation of the work in international research setting;
	Teoretisk innsikt	E	7		Dokumenterer oppgaven, og spesielt innledningen, at kandidaten har avansert kunnskap om fagfeltets teori og metoder generelt og spesialisert innsikt i et avgrenset område av spesiell betydning for oppgaven?	Report (specifically the introduction) shows advanced knowledge about theory and methods in the discipline, especially in the scope of the project
	Målbeskrivelse	E	3		Er målene og/eller aktuelle hypoteser presentert på en klar og forståelig måte?	Clear research objectives
	Eget bidrag	I	5		Har kandidaten selv generert viktige elementer/problemstillinger i oppgaven? Bruker studenten relevante ressurser (databaser m.m.) til å fremskaffe aktuell og oppdatert litteratur og bakgrunnskunnskap for arbeidet?	Generation of significant aspects & hypotheses; Use of relevant resource to find relevant state-of-the-art literature
Metoder & arbeidsform	Ferdighetsnivå	E+I	10	Methods & concepts	Behersker kandidaten relevante metoder og bruker dem i eget arbeid på en hensiktsmessig og integrert måte?	Knowledge and usage of relevant methods
	Arbeidsform	I	5		Viser kandidaten evne til planmessig og metodisk arbeid?	Show structured approach when working on the problem
	Arbeidsinnsats	I	5		Viser kandidaten evne til høy arbeidsinnsats og solid faglig engasjement?	Productivity and commitment
	Selvstendighet	I	10		Kan kandidaten arbeide og bruke relevante metoder selvstendig og gjennomføre et selvstendig forsknings - eller utviklingsprosjekt under veiledning? Vises det personlig initiativ? Hvilke typer hjelp og veiledning har kandidaten mottatt i ulike faser av arbeidet? Har kandidaten evne til å dra nytte av forskningsgruppens fagkompetanse i eget arbeid?	Independence, proactiveness, making use of the industrial economics research section's expertise in own work
Resultater & diskusjon	Arbeidet	E+I	10	Results & discussion	Viser arbeidet kreativitet og/eller bidrar til nytenkning/nyskapning? Gir arbeidet inntrykk av å være spesielt omfattende? Hvordan vurderes kvaliteten på og betydningen av ny kunnskap/nye resultater som er generert i arbeidet?	Creativity, innovation, extensiveness, quality and significance of the created knowledge / results
	Analyse og diskusjon	E	10		Er analyse, fortolkning/syntese og diskusjon faglig fundert og begrunnet og tydelig koblet til problemstillingen? Ligger diskusjonen på et høyt faglig nivå? Kan kandidaten anvende sine kunnskaper og ferdigheter på nye områder og plassere resultatene i en større sammenheng?	Are analysis, interpretation, discussion scientifically founded and connected to the research objectives? Level of the analysis? Application of knowledge and skills in new areas? Placing results in broader context
	Kritisk refleksjon	E	10		Gir kandidaten en rimelig vurdering av betydningen av resultatene? Forholder kandidaten seg kritisk til ulike informasjonskilder? Er usikkerhetsmomenter, som metodefeil, målefeil og annet vurdert og diskutert? Er relevante fag-, yrkes- og forskningsetiske problemstillinger analysert?	Critical reflection on results, information sources, used methods?
	Eget bidrag /måloppnåelse	E	5		Evner kandidaten klart å skille eget bidrag fra andres? Inneholder det skriftlige arbeidet en konklusjon der resultatene oppsummeres på en god måte med vurdering av i hvilken grad målene er nådd? Foreligger et fornuftig og begrunnet forslag til ytterligere undersøkelser eller potensialet for slike?	Seperating own contribution; conclusions relating back to research objectives; future work
Fremstilling	Struktur	E	5	Report	Har det skriftlige arbeidet en stringent oppbygning (normalt IMRaD: Introduction, Methods, Results and Discussion)? Er arbeidet generelt oversiktlig?	Structure of the report
	Språk	E	5		Kan kandidaten presentere problemstilling og resultater med nødvendig faglig presisjon? Er den godt lesbar med høy kvalitet på språket som er benyttet?	(Precision of) language, readability
	Form	E	5		Er det benyttet en enhetlig stil for referanser, figurer og tabeller? Er kvalitet på figurer og tabeller tilfredsstillende? Behersker kandidaten fagområdets uttrykksformer?	Uniform style of: References, Figures, Tables, Quality of Fig & Tab, Adequate jargon usage
TOTAL			**100**			

Selvplagiat

- Mest relevant i forbindelse med videreføring av prosjekt til masteroppgave.
  - «Gjenbruk» av tekst fra bakgrunnskapittel, litteraturstudie, problembeskrivelse, …
- Se retningslinjer fra Instituttledelsen lastet opp på Blackboard (utdrag på neste lysbilde).

Fremgangsmåter for å unngå selvplagiat

Fra **Retningslinjer vedr. selvplagiat i IØTs emner, inkludert prosjekt- og masteroppgave (10.04.2024)**

Det sentrale her er referering. Man må gjerne gjenbruke eget materiale så lenge dette gjøre eksplisitt. Grunnen til at gjenbruk må gjøres eksplisitt, er for å unngå at man får dobbel uttelling for et arbeide. Dette ligger i studieforskriftens definisjon av fusk:

> *"Fusk defineres som en opptreden i strid med NTNUs regelverk for eksamensavvikling, og som på en urettmessig måte kan føre til at kandidatens resultater blir vurdert bedre enn ellers." (§ 6-4)*

I praksis betyr det at når det settes karakter for masteroppgaven så vil omfanget som vurderes være kun det arbeidet som er gjort i det semesteret man utfører masteroppgaven. Dette er for å unngå at man får dobbel uttelling for de delene av prosjektoppgaven som man benytter i den innleverte masteroppgaven.

Det er altså lov å bruke materiale fra prosjektoppgaven i den etterfølgende masteroppgaven, så lenge det gjøres eksplisitt hva og hvor mye. En måte å gjøre dette på er å beskrive gjenbruken samlet i et forord i masteroppgaven.

Eksempel: Legg til i preface

Preface

This master's thesis concludes our Master of Science in Industrial Economics and Technology Management at the Norwegian University of Science and Technology (NTNU). The specialization of the degree is in Managerial Economics and Operations Research. The thesis is a continuation of the work conducted for our project report in the course TIØ4500 - Managerial Economics and Operations Research during the fall of 2023 (Rognes & Tronsaune, 2023).

Parts of this thesis are based on the aforementioned project report. Parts of Chapter 1, Chapter 2, and Chapter 3 are based on the corresponding chapters in the project report. Section 4.1 and Section 4.2 are revised versions of Sections 4.2 and 4.3 in the project report. Section 5.1 is based on the corresponding section from the project report, but is rewritten to incorporate the new perspectives specific to this thesis. Sections 5.2.1 and 5.2.2 are revised versions of Sections 5.2.1 and 5.2.2 from the project report. Further, Section 6.1.1 is based on the corresponding section from the project report. Parts of the content in Sections 6.3 and 6.4 in the project report are included in Section 6.1.3 and Section 6.2.1, respectively. Section 8.3 is based on Section 7.1 in the project report, except for Sections 8.3.4 to 8.3.6, which are new. Finally, Section 8.1 is based on Section 7.2 in the project report. The contributions presented in this thesis are however original to the thesis.

We thank our supervisor Magnus Stålhane, professor of Operations Research at NTNU, and co-supervisor Vibeke Hvidegaard Petersen, PhD candidate at NTNU, for their support and guidance on the thesis. We also thank Jørn Vatn, professor of Reliability, Availability, Maintainability, and Safety, for his insight on degradation processes for offshore wind turbine components. Lastly, we would like to thank Iver Bakken Sperstad, Senior Research Scientist at SINTEF, for providing valuable insight on models for synthetic weather data generation.

Eksempel: Legg til i intro til hvert kap.

Problem Description and Mathematical Model

This chapter describes the LRP this thesis aims to solve. Gulaker et al. (2022) formulate the VRP that describes the real-world routing problem Oda's solver Navegante solves for each shift. In this thesis, an extension of this model to solve the LRP is proposed: a two-stage stochastic mixed integer program, with the depot location decision in the first stage and the vehicle routing decision in the second stage. The stochastic aspect regards the stochastic customer presence, where the second stage is solved for a realization of the stochastic parameters in the first stage. The two stages of the problem are described in Section 4.1 and 4.2, respectively. A mathematical model of the problem is then presented in Section 4.3.

Background

In this chapter, we provide a general introduction to offshore wind farm installation. The content is to a large extent the same as in our earlier work, Bruu and Thorsen (2021). First, we introduce the concept and main characteristics of an offshore wind farm in Section 2.1. In Section 2.2 we provide a description of the offshore wind supply chain, with emphasis on the main offshore installation activities and concepts. Common installation vessels are further described in Section 2.3. Finally, we describe relevant aspects of vessel charter contracts and how weather affects the installation process in Section 2.4 and Section 2.5, respectively.

Eksempel: Legg til i intro til hvert kap.

Literature Review

In this chapter, we provide a review of existing literature that is relevant for the problem related to the design of charter contracts for installation vessels at offshore wind farms. Firstly, we explain the search strategy in Section 4.1. The focus of Section 4.2 is literature that directly addresses the planning problem for offshore wind farm installation, while Section 4.3 discusses literature on the more general maritime fleet size and mix problem. The content in these sections is to a large extent based on our earlier work (Bruu and Thorsen 2021), but with added consideration for solution methods. In Section 4.4 we review relevant literature on the application of B&P as a solution method for stochastic problems. Finally, we summarize the contribution of this thesis in Section 4.5.

Bruk av KI-verktøy (Hentet fra Innsida 31.08.2025)

Bruk av kunstig intelligens (KI-verktøy) i studentoppgaver

Det er tillatt å bruke KI som hjelpemiddel ved utarbeidelse av studentoppgaver. Hvis du bruker KI-verktøy kan du gjøre rede for din bruk av verktøyet enten i metodekapittelet, i et eget avsnitt eller i et skjema. Se under «Å gjøre rede for bruk av kunstig intelligens» for eksempelredegjørelse.

Riktig bruk av KI-verktøy vil som hovedprinsipp være at studenten benytter KI-verktøyet som et hjelpemiddel, men selv utarbeider besvarelsen og oppgir kilder.

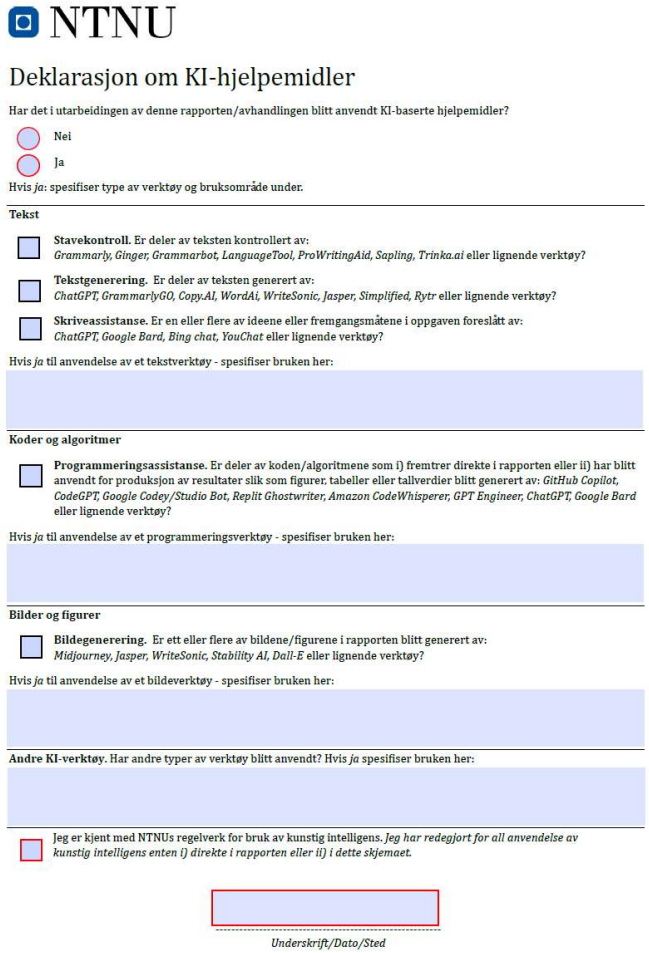
Studentoppgaver ved NTNU er skriftlige og selvstendige oppgaver. Alle oppgaver blir vurdert på samme vurderingskriterier. Dette uavhengig av om man bruker KI-verktøy som hjelpemiddel eller ikke.

Språkmodeller som ChatGPT er ikke kilder til fakta, og kan ikke siteres for faktaopplysninger. Bruk originale kilder.

https://i.ntnu.no/wiki/-/wiki/Norsk/Kunstig+intelligens+i+bachelor-+og+masteroppgaver (31.08.2025)

KI-deklarasjonsskjema

- Innfylt KI-deklarasjonsskjema lastes opp på Inspera i forbindelse med innlevering av prosjekt- og masteroppgave.
- Skjemaet finnes på Innsida.



NTNU

Deklarasjon om KI-hjelpemidler

Har det i utarbeidingen av denne rapporten/avhandlingen blitt anvendt KI-baserte hjelpemidler?

Nei

Ja

Hvis *ja*: spesifiser type av verktøy og bruksområde under.

Tekst

Stavekontroll. Er deler av teksten kontrollert av:
Grammarly, Ginger, Grammarbot, LanguageTool, ProWritingAid, Sapling, Trinka.ai eller lignende verktøy?

Tekstgenerering. Er deler av teksten generert av:
ChatGPT, GrammarlyGO, Copy.AI, WordAi, WriteSonic, Jasper, Simplified, Rytr eller lignende verktøy?

Skriveassistanse. Er en eller flere av ideene eller fremgangsmåtene i oppgaven foreslått av:
ChatGPT, Google Bard, Bing chat, YouChat eller lignende verktøy?

Hvis *ja* til anvendelse av et tekstverktøy - spesifiser bruken her:

Koder og algoritmer

Programmeringsassistanse. Er deler av koden/algoritmene som i) fremtrer direkte i rapporten eller ii) har blitt anvendt for produksjon av resultater slik som figurer, tabeller eller tallverdier blitt generert av: *GitHub Copilot, CodeGPT, Google Codey/Studio Bot, Replit Ghostwriter, Amazon CodeWhisperer, GPT Engineer, ChatGPT, Google Bard* eller lignende verktøy?

Hvis *ja* til anvendelse av et programmeringsverktøy - spesifiser bruken her:

Bilder og figurer

Bildegenerering. Er ett eller flere av bildene/figurene i rapporten blitt generert av:
Midjourney, Jasper, WriteSonic, Stability AI, Dall-E eller lignende verktøy?

Hvis *ja* til anvendelse av et bildeverktøy - spesifiser bruken her:

Andre KI-verktøy. Har andre typer av verktøy blitt anvendt? Hvis *ja* spesifiser bruken her:

Jeg er kjent med NTNUs regelverk for bruk av kunstig intelligens. *Jeg har redegjort for all anvendelse av kunstig intelligens enten i) direkte i rapporten eller ii) i dette skjemaet.*

Underskrift/Dato/Sted

Bruk av KI-verktøy i prosjekt- og masteroppgaver

Diskuter i grupper

- Hvilke (generative) KI-verktøy har dere erfaring med fra tidligere emner eller jobb?
- Hvilke verktøy kan det være mulig å benytte i prosjekt- og masteroppgave? Og, hvordan? I hvilke deler av arbeidet?
- Hvilke utfordringer kan det være ved å bruke disse verktøyene?

Oppsummering

- Hvorfor er rapportskriving viktig
- Struktur: IMRaD
- Innhold: Hva skrives hvor
- Råd om stil og språk
- Selvplagiat
- Bruk av KI